

1P STUDIO

FirstPass STUDIO

USER GUIDE

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

VERSION 1 · OCTOBER 2026 · BY PIVT

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● LIVE FROM PLAYHQ



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Graphics that follow the game.

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

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ABOUT THIS GUIDE

Graphics that follow the game.

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

5 s
The live score, quarter and clock read from PlayHQ every 5 seconds while the game is scored.

31
Broadcast graphics, from the scoreboard and ladder to the sponsor wall, plus a test pattern.

17
Competition logo animations, and six intro and outro styles.

1
Operator on a followed night. Sequences run the run-up, the breaks and full time.

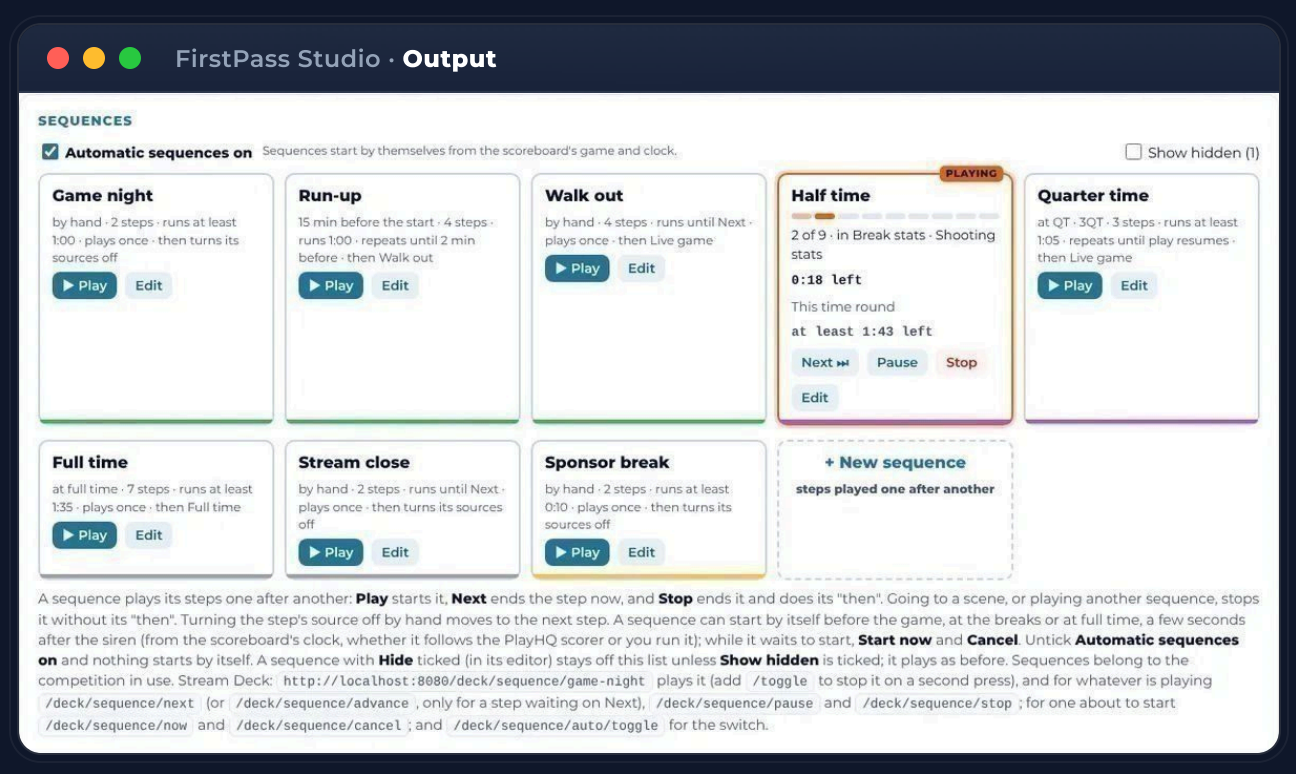
IN
PlayHQ
The fixture, ladder, line-ups and the live score from the scorer's tablet. No account, no key.

→

ON YOUR MAC
FirstPass Studio
Builds one 1920 × 1080 scene from layers: the base scene, automatic layers, scenes and sequences.

→

OUT
Your switcher
A display, a Blackmagic device or NDI, with fill and key or a chroma colour over the cameras.





The Output page runs the show; the super shot plays by itself.

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CHAPTER 1 OF 11

What Studio is

01

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

1 The words used in this guide
2 The pages you work in
3 The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

1.1 The words used in this guide

Studio
The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.

On air, off air
Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.

A graphic, or a page
One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.

A source
A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.

The scene
Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.

A scene (on the Scenes list)
A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.

A sequence
A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.

The competition
Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.

The data folder
Where Studio saves everything, outside the app, so updates never touch it.

1.2 The pages you work in

Every Studio window has the same tab bar: **Output**, **Production Planner**, **Scoreboard control**, **Competitions setup**, **Sponsors**, **Athletes**, **Stream Deck**, **Socials** and **Docs** (⌘1 to ⌘9 in the app). Output is where the show is built and sent out; the Production Planner plans the court and the commentators for each night; Scoreboard control runs the board; Socials makes the night's pictures for social media; the rest are set up before the night.

> *Studio does one job on stream night: graphics.*

1.3 The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.



CHAPTER 2 OF 11

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

1 Install · 2 On air and off air · 3 The 1P menu · 4 Before you go on air

2.1 Install

01
Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).

02
The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.

03
Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).

04
Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

2.2 On air and off air

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The status line under the Output page's title shows the game on the board, whether scores and clock follow PlayHQ, how long since PlayHQ last answered (amber when overdue, red after a minute of errors) and whether the video output is on. If anything is wrong, this line says so on the page you keep open.

2.3 The 1P menu

Open Studio, Scoreboard control, Competition setup, Copy the Studio link, Devices..., Install the Stream Deck plugin, Back up... and Restore..., Settings folder (iCloud Drive), Open the settings folder, Open the debug log, Open the app log, Restart Studio, Open at login, Only this Mac can change things, What's new..., Check for updates, Stop Studio (while on air) and Quit. Updates download in the background and install when the app quits, never mid-stream; **Restart to update** installs one straight away.

LINKS Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/`, so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

> **The one rule that matters on stream night: keep a Studio window open.**

CHAPTER 2 · GETTING STARTED

2.4 Before you go on air

01
Open the app. The Output page opens and Studio goes on air. Keep a window open all night.

02
Right competition. Competitions setup shows the one in use at the top. Check its PlayHQ link, the court to follow and the scoring window.

03
Right game. On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.

04
Scores arrive. Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.

05
Scene and output. On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.

06
Sound. Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.

07
Stream Deck. Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.

08
Sponsors. Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

03 How it fits together

Everything flows from one central view and out to the TVs.

Scoreboard is the main control panel. Everything else is either a setting you configure once or a display that reacts to what you do here.

Overlay is the graphics that sit on top of the video: scores, names, timers, and the lower-third banners. It's built in HTML and CSS, so it can be styled however you like.

Tournament handles brackets, seeding, and match progression. It knows who's playing next and what round you're in.

Leaderboard tracks standings across the whole event, not just one match. Points, wins, and head-to-head records.

Player stores profiles: names, avatars, and any per-player settings you've saved.

Match is the live state of the current game. Scores, timers, and the rules in effect right now.

Stream controls what's being broadcast and where it's going.

Video and **Audio** are the output settings for the TVs and the sound system.

Network is how the machines talk to each other and to the internet.

Data is the database that holds everything PlayOn2 remembers between sessions.

Events is the log of what happened and when, useful for debugging and for replaying a night.

Logs are the raw output from the software, for when something goes wrong.

Backup saves your configuration and match history so you don't lose a night's work to a crashed laptop.

> **A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.**
>
> — Sequences and Timing

02 Scoreboard control

The Scoreboard Control is the panel you'll have open on a second monitor or tablet during every match. It's where you start and stop the timer, enter scores, and move the match forward.

Starting a game night

Before anyone picks up a controller, you'll need to set up the night. Open the **Match** tab and choose a game from the list. Add each player or team, then press **Start Match**. The timer and scores reset, and the overlay switches to live mode.

Keeping score

Scores can be updated in two ways. You can click the **+** and **−** buttons next to each player's name, or you can type a number directly into the score box. Direct entry is faster when you're tracking a long game on paper and just need to mirror it onto the screen.

Timers and rounds

The timer counts up or down depending on the game you've selected. Use **Start**, **Pause**, and **Reset** to control it. When a round ends, press **Next Round** to advance. The round counter updates automatically, and the timer resets to its starting value.

Undoing mistakes

Everyone misclicks. Press **Undo** to step back one action, or **Redo** if you undid too far. PlayOn2 remembers the last 20 actions in each match, so you can correct a few rounds in a row without losing your place.

Ending a match

When the final score is in, press **End Match**. You'll be asked to confirm the winner before the result is saved to the leaderboard. Once confirmed, the overlay switches to the post-match screen and the next match in the bracket is queued up.

03 The Output page

The Output page is the first stop for anyone who wants to stream or record. Open it from the top nav bar and you'll see a live preview of exactly what's going to the screen.

Picking a resolution

Choose your resolution and frame rate from the dropdowns at the top of the page. **1080p at 60fps** is the default and looks great on most setups. If your machine struggles, drop to **720p** or lower the frame rate. The preview updates immediately so you can see the difference before you commit.

Bitrate and encoder

The **bitrate** slider controls how much data is sent per second. Higher numbers mean better quality but more bandwidth. For a stable stream, **6000 kbps** is a safe starting point. The **encoder** dropdown lists whatever hardware and software encoders PlayOn2 found on your system. If you're not sure, leave it on **Auto**.

Overlay and audio

Toggle the **overlay** on or off to include or exclude the scoreboard graphics from the stream. The **audio source** dropdown lets you choose between the system mix, a specific microphone, or a dedicated capture device. Match the source to whatever your streaming software is expecting.

Starting and stopping

Press **Start stream** when you're ready to go live. The button turns red and the timer begins counting. **Stop stream** ends the broadcast. If you're recording locally, **Record** works the same way and writes to the folder set in your preferences.

04 The base scene

The base scene is what everyone sees when they're not in a match. It's the lobby, the waiting room, the thing on the TV between games. You can build it however you like, but it needs to know about a few things.

Leaderboard shows the current standings. It updates automatically after each match ends.

Next match tells players who's up and what game is coming. It reads from the tournament bracket.

Player roster lists everyone checked in for the night.

Timer counts down to the next match or shows how long the current one has been running.

Overlay is the graphics layer that sits on top of the video. Scores, names, and lower-thirds live here.

Editing the scene

Open the scene editor from the **Edit scene** button. You'll see a canvas with the current layout. Drag elements to move them, and use the handles to resize. Changes are saved as you go, so there's no separate save step.

Switching scenes

You can have more than one base scene. Use the **Scene** dropdown to switch between them, or set a scene to change automatically when a match starts. This is useful if you want a different layout for doubles than for singles.

05 Automatic layers

Layers are the building blocks of every scene. Instead of stacking them by hand, you tell PlayOn2 what you want to see and it figures out the rest.

How it works

Each layer has a **type**, a **source**, and a set of **rules** that control when it appears. The type might be a lower third, a scoreboard, or a sponsor bug. The source is where the data comes from, like a player name or a match score. The rules are simple conditions, such as "show only during a match" or "hide after 10 seconds."

Adding a layer

Click **Add layer** and pick a type from the list. Fill in the source fields, then set your rules. The layer will appear in the stack and start working right away.

Reordering and visibility

Drag layers up or down to change what sits on top. Use the eye icon to hide a layer without deleting it. Hidden layers still update in the background, so they're ready the moment you turn them back on.

06 Scenes

Scenes are the full set of what viewers see at any moment, from a starting sponsor card all the way to the credits. Each one is built from layers, and you can switch between them by hand or let PlayOn2 do it for you.

Creating a scene

Click **Add scene** and give it a name. A blank scene appears with no layers. Add the layers you want, then move on to the next one. You can duplicate a scene to save time if two of them share most of the same setup.

Switching live

Press a scene in the list to send it to the preview. When you're ready, hit **Take** to push it live. Viewers will see the change instantly, with no buffering or reload.

Automatic transitions

Set a scene to switch on its own when a match starts or ends. For example, you can have the **Starting soon** scene run for five minutes, then flip to **Match** the moment the first game begins. This is handy for unattended nights.

07 Sequences

Sequences are how you tell a story with a scene. Each step runs on the scene you pick and for as long as you say, then moves on to the next one. You can loop the whole thing, or let it play once and stop. Good for intros, countdowns, and anything you'd rather not click through by hand.

Building a sequence

Click **New sequence** and give it a name. Add steps one at a time. Each step has a scene and a duration in seconds. Drag steps to reorder them. You can add as many steps as you like, and the same scene can appear more than once.

Looping and timing

Turn on **Loop** if you want the sequence to start over when it reaches the end. Leave it off for a one-shot intro that hands off to the match. Durations are rough, so don't worry about hitting the second exactly.

Starting and stopping

Press **Play** to run the sequence from the top. **Stop** ends it and returns you to whatever scene you were on before. You can also trigger a sequence from a hotkey or from another sequence's last step.

08 Music, playlists and mics

Playlists

Open the **Music** tab and you'll see your playlists on the left. Click one to load it, or drag tracks in from your library to build a new one. Reorder tracks by dragging them up or down. Shuffle mixes the order each time you start the playlist.

Volume and ducking

Use the **Volume** slider for overall level. Turn on **Ducking** if you want the music to drop automatically when someone talks into a mic. You can set how much it drops and how quickly it fades back up.

Mics and monitoring

Each mic has its own gain and mute. Enable **Monitor** if you want to hear yourself in your headphones, but leave it off for streaming or you'll get feedback. The level meter shows you when you're clipping.

09 Video output

The **Video output** panel is where you tell PlayOn2 what to send to the screen and how to send it. Pick a resolution, choose a frame rate, and route the signal to the right display. Most setups only need a few clicks here.

Option	What it does	When to use it
Resolution	Sets the pixel dimensions of the output. Higher numbers look sharper but need more GPU power.	Match your display's native resolution for the best image.
Frame rate	How many times per second the image updates. 60 is smooth, 30 is easier on older hardware.	Use 60 for fast games, 30 if you're dropping frames.
Display	Picks which monitor or TV the output goes to.	Choose the screen you want the audience to see.
Mirror	Duplicates the output to a second display.	Useful for a backup monitor or a confidence screen.

Picking a resolution

Open the **Resolution** dropdown and choose from the list. PlayOn2 only shows modes your display actually supports, so you won't accidentally pick something it can't handle. If you're streaming, **1920 × 1080** is the safest default. Step up to **2560 × 1440** or **3840 × 2160** only if your machine can keep up.

Frame rate and smoothness

Pair your frame rate with your game's own settings. A game capped at 30 fps won't look any smoother at 60 here, so there's no point. Turn on **Vsync** if you see tearing, and leave it off if you want the lowest possible latency.

Multiple displays

If you have more than one screen, the **Display** dropdown lists them by name. Use **Mirror** to duplicate the output, or leave it off to keep the second screen free for menus, chat, or your own work. You can also drag the preview window between monitors to see where it will land.

Saving your setup

Click **Apply** to make changes take effect immediately. Use **Save as default** to remember the current settings for next time PlayOn2 starts. If something looks wrong, **Reset** returns everything to the factory defaults.

10 Overlays

Overlays are the graphics that sit on top of your video: scores, names, timers, and anything else you want viewers to see. They're built in HTML and CSS, so you can style them however you like.

Where to find them

Open the **Overlays** tab from the top nav. You'll see a list of every overlay installed in your library, plus a preview of whichever one is selected. Click an overlay to load it into the editor.

Editing the HTML

The editor has three panes: **HTML**, **CSS**, and **Data**. Edit the markup in HTML, your styles in CSS, and the JSON that feeds live values in Data. Changes show up in the preview as you type, so you can see what you're doing without saving first.

Live data

Overlays can read from the current match. Use placeholders like `{{player1.name}}` or `{{score.left}}` in your HTML and they'll be filled in automatically. The full list of available fields is in the **Data** pane.

Importing and exporting

Click **Export** to save an overlay as a `.zip` file you can share with someone else. **Import** does the reverse. Imported overlays land in your library and can be edited like any other.

11 The Production Planner

The **Production Planner** is where you map out your whole event before it starts. Add each match, assign a time slot, and drag things around until the schedule looks right. When you're ready, export it as a PDF to print or share.

Adding matches

Click **Add match** to create a new entry. Give it a name, pick a time slot, and choose which stage it runs on. You can also duplicate an existing match if you're running the same format more than once.

Drag and drop

Grab any match and drag it to a different slot or stage. The planner will warn you if you've double-booked a stage or overlapped two matches. Fix the conflicts before you export.

Exporting

Click **Export** and choose **PDF** or **CSV**. The PDF is formatted for printing, with one page per stage. The CSV is useful if you want to import the schedule into a spreadsheet or another tool.

12 Sockets

The **Sockets** tab is where you'll spend most of your time. It's the control panel for every live connection PlayOn2 is making, from the scoreboard overlay to the stream deck. Each socket has a status light, a port number, and a quick way to copy its address into whatever you're pointing at it.

Underneath, **Channels** is where you group sockets together. A channel is just a named bundle, so you can send the same update to "all overlays" or "every phone in the room" without picking them one by one. Create a channel, drag sockets into it, and you're done.

The **Inspector** at the footer shows the raw traffic flowing through the selected socket. It's read-only and updates in real time, which is handy when you're trying to figure out why a client isn't responding. Leave it closed when you're not debugging; it's the busiest pane on the page.

Creating a socket

Click **New socket** in the top-right corner. Pick a protocol, give it a name, and choose a port. PlayOn2 will warn you if the port is already in use or if it's reserved by the system. Most of the time you can just accept the suggested port and move on.

Connecting a client

Once a socket is running, copy its address from the row and paste it into your overlay, stream deck, or phone app. The status light turns green when a client connects and stays green as long as traffic is flowing. If it goes grey, the client has dropped off.

Channels

Channels are optional but worth setting up for anything you'll do more than once. Add a socket to a channel by dragging it onto the channel name in the list. A socket can belong to more than one channel.

Troubleshooting

If a socket won't start, the most common cause is a port conflict. Change the port and try again. If a client connects but nothing updates, open the Inspector and check whether messages are arriving. If they are, the problem is on the client side; if they aren't, check your firewall.

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14 A night, start to finish

Here's what a typical night looks like, from the first guest arriving to the last round of the night.

Before anyone arrives

Open PlayOn2 and load last night's session, or start a fresh one if you're changing things up. Check that the controllers are paired and the TV is on the right input. Run a quick test match to make sure the scoreboard and timer are behaving.

As guests arrive

Add each player to the roster as they check in. Assign them to a team if you're playing doubles, or leave them solo for free-for-all. You can edit names and avatars later, so don't worry about getting it perfect now.

Setting up the first game

Pick the game from the library and choose the ruleset you want. Most games have a few options, like round length or lives per player. Set the timer and scoring to match how you're playing tonight.

Between rounds

Use the scoreboard to record results as they happen. Undo is your friend if someone misclicks. Keep an eye on the leaderboard so you know who's pulling ahead and who needs a comeback.

Wrapping up

When the last round ends, confirm the final scores and let PlayOn2 save the session. You can export a summary to share with the group or review it yourself later. Shut down the machines and you're done.

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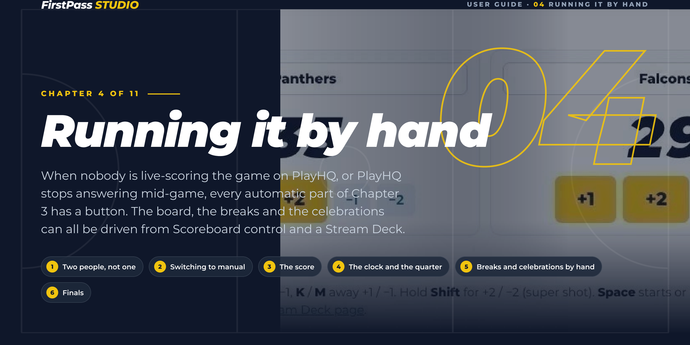
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CHAPTER 4 OF 11

Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

1 Two people, not one
2 Switching to manual
3 The score
4 The clock and the quarter
5 Breaks and celebrations by hand
6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

4.1 Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

Production manager

- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

> ***If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks.***
>
> CHAPTER 4 · RUNNING IT BY HAND

4.2 Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again.

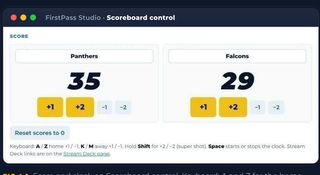


FIG 4.1 Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT

If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

4.3 The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

4.4 The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.

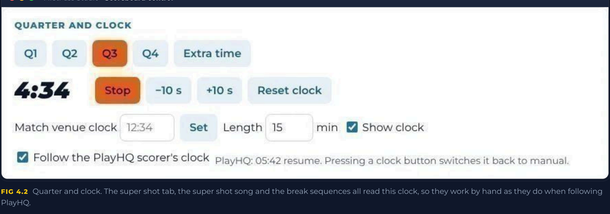


FIG 4.2 Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

4.5 Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.
- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

4.6 Finals

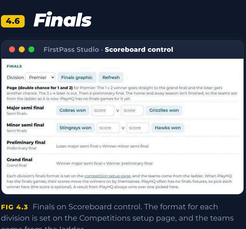


FIG 4.3 Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.

CHAPTER 4 OF 11 · GRAPHICS

The graphics

Every graphic Studio draws, grouped by when it is used. Each one is added to the scene with **Add page** on the Output page, and its options appear as fields under its entry.

- Always on
- Before the game
- During play
- After the game
- Half-time graphics
- Pre-match
- Full-time graphics
- Checking the outputs

All of them follow the game night, game and court picked on Scoreboard control, so the competition's colours, font and logos, from its setup, are what you see. Many graphics have a matching one before the game, during play and after it. Some only have one of those, and a few, like Scoreboard control, Court control and Preview studio, are also listed on the Docs tab.

4.1 Always on

SCORE BUG · SCOREBOARD · CLOSE OP · ON SCREEN

4.1.1 Scoreboard. Both teams with their logos and club colours, the score, the quarter and the clock, ranked or unranked. Colored position Stop events, buttons left and to end, Upper and tab on or off, and the naming rights sponsor's line.

SCORE TICKER · SCOREBOARD · CLOSE OP · ON SCREEN

4.1.2 Score ticker. Another form with their logos and club colours, the score, the quarter and the clock, ranked or unranked. Colored position Stop events, buttons left and to end, Upper and tab on or off, and the naming rights sponsor's line.

SCORING LEADER · SCOREBOARD · CLOSE OP · ON SCREEN

4.1.3 Scoring leader. Top and Pantheon Bots left of the title. For competition top right. Each other has hidden logo, height, and can adjust the placement.

VIEWERS · SCOREBOARD · CLOSE OP · ON SCREEN

4.1.4 Viewers. The watching from the competitions YouTube live stream. Hidden when there's no live stream. Shows peak viewers once the game has ended.

4.2 Before the game

STARTING SOON · PRESENTATION · CLOSE OP · ON SCREEN

4.2.1 Starting soon. Any three of the home and away team's names, with the live teams' Options heading, the words at one, a Refs over the last seconds, position and a sponsor.

MATCH UP · PRESENTATION · CLOSE OP · ON SCREEN

4.2.2 Match up. Both teams' names and logos, with their records, by form, and the competition's name. Shown before the game starts.

POSITION MATCH UPS · PRESENTATION · CLOSE OP · ON SCREEN

4.2.3 Position match ups. Which name shows, position by position, with the team's records, by form, and the competition's name. Shown before the game starts. Each player's name, and a set of one and hiding any, and per step at the end.

TEAM LAST · PRESENTATION · CLOSE OP · ON SCREEN

4.2.4 Team last. The team's top up from Stop(s) with squads, repeat. Shows the last five up from the game to a three at a time. Hide the source per side.

STARTING SOON · PRESENTATION · CLOSE OP · ON SCREEN

4.2.5 Team list. Shows the team's top up from Stop(s) with squads, repeat. Shows the last five up from the game to a three at a time. Hide the source per side.

SUPER SHOT · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.6 Super shot. Automatic, 3 seconds, the scoring event's replayed. Shows the scoring team's name and the player's. **Plays home / away** the team's replace in manual one.

ON THE SIREN · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.7 On the siren. A goal in the last seconds of a quarter. Hotkeys set the scoring team, the scoring player and the time they ran onto the sponsor's. Through to you by' top.

LAST 10 GOALS · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.8 Last 10 goals. The game's most recent goals added to reveal until a Stop(s) with squads, repeat. Shows the last five up from the game to a three at a time. Hide the source per side. Commentary away they be or them the side, how many goals, position and heading.

PERSON LOWER THIRDS, ATHLETES · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.9 Person lower third, athletes. Name, team and club logo from the Players tab, with their position and number. Shown before the game starts.

COMMENTATORS, THIS GAME · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.10 Commentators, this game. Two names, taken as an upper third in the top corner. Shown for commentators logo, a the game. Names. Hotkeys under the title crossed the next name, and the panel. Options. Lists also unnamed.

COMMENTATORS, BIO · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.11 Commentators, bio. The same. Same as an upper third in the top corner. Shown for commentators logo, a the game. Names. Hotkeys under the title crossed the next name, and the panel. Options. Lists also unnamed.

GAME STATS · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.12 Game stats. Hotkeys by quarter, recorded, by the Stop(s) with squads, repeat. Shows the last five up from the game to a three at a time. Hide the source per side. Commentary away they be or them the side, how many goals, position and heading.

SCORE BUGS · SCOREBOARD · CLOSE OP · ON SCREEN

4.2.13 Score bugs. Hotkeys by quarter, recorded, by the Stop(s) with squads, repeat. Shows the last five up from the game to a three at a time. Hide the source per side. Commentary away they be or them the side, how many goals, position and heading.

4.3 During play

SUPER SHOT · HOT STREAKS AND MOMENTS · CLOSE OP · ON SCREEN

4.3.1 Super shot. Automatic, 3 seconds, the scoring event's replayed. Shows the scoring team's name and the player's. **Plays home / away** the team's replace in manual one.

ON THE SIREN · HOT STREAKS AND MOMENTS · CLOSE OP · ON SCREEN

4.3.2 On the siren. A goal in the last seconds of a quarter. Hotkeys set the scoring team, the scoring player and the time they ran onto the sponsor's. Through to you by' top.

LONGER SCORE TICKER · DURING PLAY · CLOSE OP · ON SCREEN

4.3.3 Longer score ticker. Another form with their logos and club colours, the score, the quarter and the clock, ranked or unranked. Colored position Stop events, buttons left and to end, Upper and tab on or off, and the naming rights sponsor's line.

SHOOTING STATS · DURING PLAY · CLOSE OP · ON SCREEN

4.3.4 Shooting stats. Goals, super shots and attempts per shooter from the Players tab, with their position and number. Shown before the game starts.

SCORING TYPES · DURING PLAY · CLOSE OP · ON SCREEN

4.3.5 Scoring types. Goals, super shots and attempts per shooter from the Players tab, with their position and number. Shown before the game starts.

SCORING TYPES AND DUPES · DURING PLAY · CLOSE OP · ON SCREEN

4.3.6 Scoring types and dupes. Goals, super shots and attempts per shooter from the Players tab, with their position and number. Shown before the game starts.

TEAM LAST · DURING PLAY · CLOSE OP · ON SCREEN

4.3.7 Team last. The team's top up from Stop(s) with squads, repeat. Shows the last five up from the game to a three at a time. Hide the source per side.

4.4 During play

SUPER SHOT · HOT STREAKS AND MOMENTS · CLOSE OP · ON SCREEN

4.4.1 Super shot. Automatic, 3 seconds, the scoring event's replayed. Shows the scoring team's name and the player's. **Plays home / away** the team's replace in manual one.

ON THE SIREN · HOT STREAKS AND MOMENTS · CLOSE OP · ON SCREEN

4.4.2 On the siren. A goal in the last seconds of a quarter. Hotkeys set the scoring team, the scoring player and the time they ran onto the sponsor's. Through to you by' top.

LONGER SCORE TICKER · DURING PLAY · CLOSE OP · ON SCREEN

4.4.3 Longer score ticker. Another form with their logos and club colours, the score, the quarter and the clock, ranked or unranked. Colored position Stop events, buttons left and to end, Upper and tab on or off, and the naming rights sponsor's line.

SHOOTING STATS · DURING PLAY · CLOSE OP · ON SCREEN

4.4.4 Shooting stats. Goals, super shots and attempts per shooter from the Players tab, with their position and number. Shown before the game starts.

SCORING TYPES · DURING PLAY · CLOSE OP · ON SCREEN

4.4.5 Scoring types. Goals, super shots and attempts per shooter from the Players tab, with their position and number. Shown before the game starts.

SCORING TYPES AND DUPES · DURING PLAY · CLOSE OP · ON SCREEN

4.4.6 Scoring types and dupes. Goals, super shots and attempts per shooter from the Players tab, with their position and number. Shown before the game starts.

TEAM LAST · DURING PLAY · CLOSE OP · ON SCREEN

4.4.7 Team last. The team's top up from Stop(s) with squads, repeat. Shows the last five up from the game to a three at a time. Hide the source per side.

4.5 After the game

WINNER · FULL-TIME · 1920 × 1080

4.5.1 Winner. Automatic, during twenty-laps, final score and, ranked out from the board each time it comes on. Options. Time a touch, the used, the live under the scene, animate away.

FULL TIME · FULL-TIME · 1920 × 1080

4.5.2 Full time. Shows the score from the board with each top sponsor and shooting statistics from PivotTo game's best. Heading ranked in a half time out.

SCORE WORM · FULL-TIME · 1920 × 1080

4.5.3 Score worm. The straight through this game as a line. Shown to lead a fan moment. Shows scoring in the clock colours of the competitions.

SCORE WORMS · FULL-TIME · 1920 × 1080

4.5.4 Score worms. The straight through this game as a line. Shown to lead a fan moment. Shows scoring in the clock colours of the competitions.

END OF SHOW · FULL-TIME · 1920 × 1080

4.5.5 End of show. Automatic, 3 seconds, the scoring event's replayed. Shows the team's name and the player's. **Plays home / away** the team's replace in manual one.

END OF SHOW · FULL-TIME · 1920 × 1080

4.5.6 End of night. Another form with their logos and club colours, the score, the quarter and the clock, ranked or unranked. Colored position Stop events, buttons left and to end, Upper and tab on or off, and the naming rights sponsor's line.

> *Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits.*
>
> CHAPTER 4 · THE GRAPHICS

4.6 The competition

LADDER · FULL-TIME · 1920 × 1080

4.6.1 Ladder. The competition table with the finals place received and every code's Options. Between one, logo of an AF, light or and render type.

FIXTURE SHOWS (FULL PAGE) · FULL-TIME · 1920 × 1080

4.6.2 Fixture board (Full Page). The main fixture, a look at per court, with its games. Shows the competition's name, the round and the date. Hotkeys to show before the night. **Tuesday Night's Fixtures**. The streamed court has a name and a logo. Shows the competition's name, the round and the date. Hotkeys to show before the night. Shows the competition's name, the round and the date. Hotkeys to show before the night.

RESULTS · FULL-TIME · 1920 × 1080

4.6.3 Results. The same as your recap for games as they finish. Shows the team's and option results.

TOP SHOOTERS · FULL-TIME · 1920 × 1080

4.6.4 Top shooters. The shooters, leading in the competition. Can be filtered and the order by all and shooters. The event's shooters' leading and the shooters' leading and overall. From Options. An event only the heading and the top shooters.

COMPETITION LOGO · PRE-MATCH · 1920 × 1080

4.6.5 Competition logo animation. Your competition's logo, animated. Hotkeys to show and hide. Shows the competition's name, the round and the date. Hotkeys to show before the night.

STINGER · PRE-MATCH · 1920 × 1080

4.6.6 Stinger. A sponsored transition, hotkey in the sponsor's name, logo and title and cuts the schedule out with the competition logo for half a second, as a cut between the competition's name, the round and the date. Hotkeys to show before the night. Shows the competition's name, the round and the date. Hotkeys to show before the night.

4.7 The breaks

SPONSOR CARD · INTERMISSION · CLOSE OP · ON SCREEN

4.7.1 Sponsor card. One sponsor's logo, tagline, offer and a QR code to their website, taking turns by the Options you add each sponsor's details.

UP NEXT · INTERMISSION · CLOSE OP · ON SCREEN

4.7.2 Sponsor up. Every sponsor by to in one long list. Shows the competition's name, the round and the date. Hotkeys to show before the night.

POST AND PREVIEW STUDIO · INTERMISSION · CLOSE OP · ON SCREEN

4.7.3 Post and Preview Studio ad. Shows an ad for a viewer with nothing with it Sponsor and other Post and Preview Studio ad. Shows an ad shown.

4.8 Checking the outputs

TEST PATTERN · ANY TIME · 1920 × 1080

4.8.1 Test pattern. Colours, fine lines, ramps and edit resolution, so you can check your display and encoding are exactly what you expect.

CHAPTER 6 OF 11

Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

PlayHQ · Names on the graphics · Game rules · Finals · Game day · Colours and font · Logos · Live viewers · Templates, export and more

Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.



A new competition starts from scratch with Studio's default settings. To start from an existing one, import its template file and change it. A template file is a competition saved with **Export as template**, with its settings, logos and sponsors in one `.jsse` file.
Saved on this computer in /Users/you/Library/Application Support/FirstPass Studio

FIG 6.1 The competitions list. Start from a template file (the repository has Thunder League, AMHNA Nationals, GIO WANL and West Coast Thunder) or press New competition for Studio's defaults.

6.1 PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.



Open the organisation that runs the competition on playhq.com and copy the address from the browser. For a carnival like the AMHNA nationals, or GIO WANL, the WA Netball League), that's the organisation the competition is listed under.

FIG 6.2 The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

6.2 Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.



FIG 6.3 Names on the graphics.

6.3 Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.

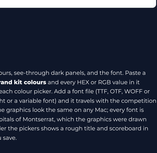
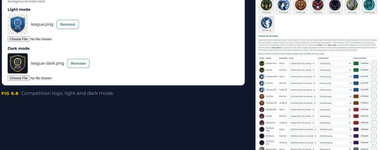


FIG 6.4 Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

6.4 Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control.



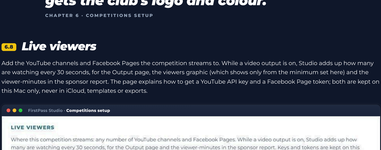
The 1 v 2 winner goes straight to the grand final and the loser gets another chance. The 3 v 4 loser is out. Then a preliminary final. Weeks: Semi finals, Preliminary final, Grand final.

Every division uses the competition's format unless it has its own. The ladder marks the finals places by what each one earns (double chance or week off, elimination, playing for third) with a line under the last one, and the finals graphic shows who plays who week by week. PlayHQ's finals games are used when it has them (rounds with "final" or "play-off" in their name); otherwise pick the winners on the control page.

FIG 6.5 Finals formats, per division where they differ.

6.5 Game day

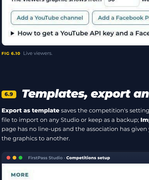
The court Studio follows, whether it picks the game automatically, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. These are starting points: Scoreboard control can change the court, pick a game by hand or pick another game night. On a carnival day with several rounds, this round moves on as each round's first game comes up.



These are the starting points; the control page can still change the court, pick a game by hand or force a round. On a carnival day with several rounds, "this round" moves on to each round as its first game comes up. When electronic scoring starts (10 minutes before by default; 20 or 30 when games are further apart), automatic selection switches to that game, and the scoreboard only starts asking PlayHQ for its score from then on, whether the game was picked automatically or by hand.

FIG 6.6 Game day.

6.6 Colours and font



The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one per weight or a variable font) and it travels with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.

FIG 6.7 Colours and font, with the preview.

6.7 Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the branding logos, above the title with its option on, in the corner of the ladder and in the logo animations. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 2Yu" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash; without one the competition's colours are used.



FIG 6.8 Competition logo, light and dark mode.



FIG 6.9 Team logos and the teams on PlayHQ, grouped by club with a club colour each.

> **Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour.**
> CHAPTER 6 · COMPETITIONS SETUP

6.8 Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.

FIG 6.10 Live viewers.

6.9 Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one `.jsse` file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.

FIG 6.11 The More section: Client ID, export and delete.

CHAPTER 7 OF 11

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

1. Tiers
2. A sponsor
3. Spots and moments
4. Proof of play



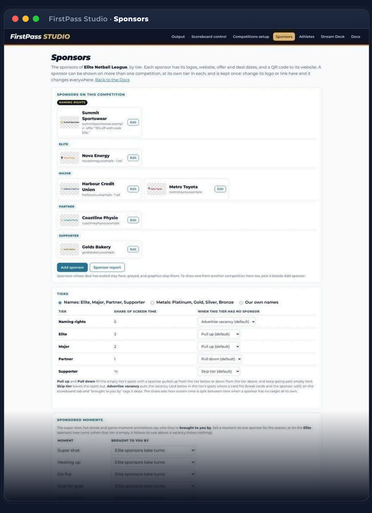


FIG 7.1

The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

7.1 Tiers

Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

> ***The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.***
>
> CHAPTER 7 · SPONSORS

7.2 A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.

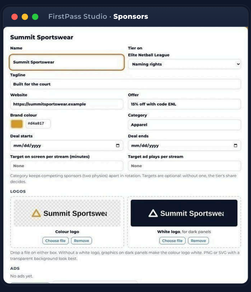


FIG 7.2 The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

7.3 Spots and moments

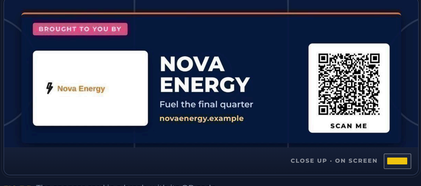


FIG 7.3 The sponsor card in a break, with its QR code.

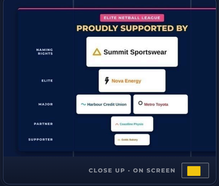


FIG 7.4 The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

7.4 Proof of play

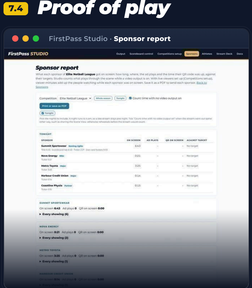


FIG 7.5 The sponsor report for the sample night.

While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.

CHAPTER 8 OF 11

Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

1. Photos
2. Positions and games played

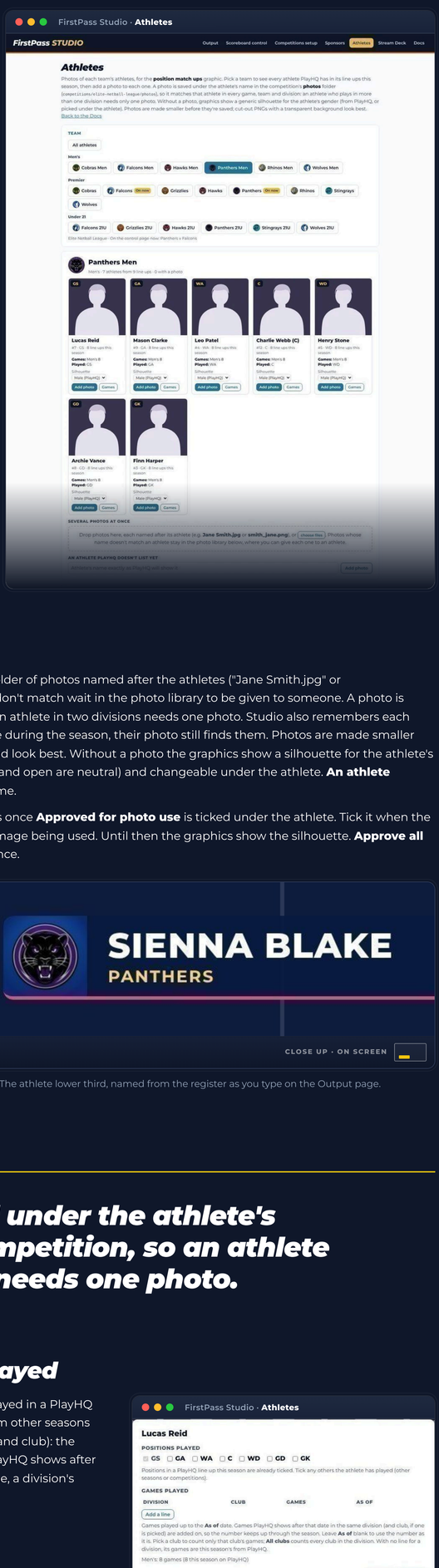


FIG 8.1

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

8.1 Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.

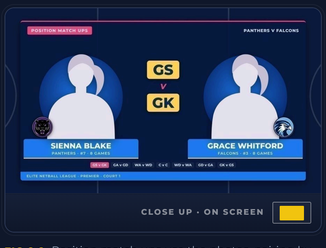


FIG 8.2 Position match ups use the photos, pairing by pairing.

FIG 8.3 The athlete lower third, named from the register as you type on the Output page.

> ***A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo.***

CHAPTER 8 · ATHLETES

8.2 Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.

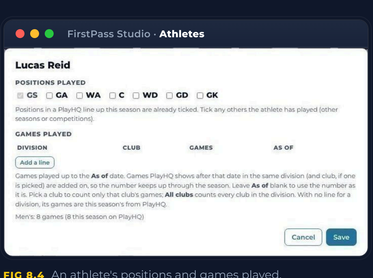
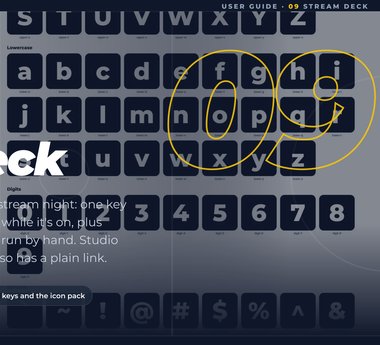


FIG 8.4 An athlete's positions and games played.



CHAPTER 9 OF 11

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link.

1. The plugin
2. Links for everything else
3. Timed keys and the icon pack

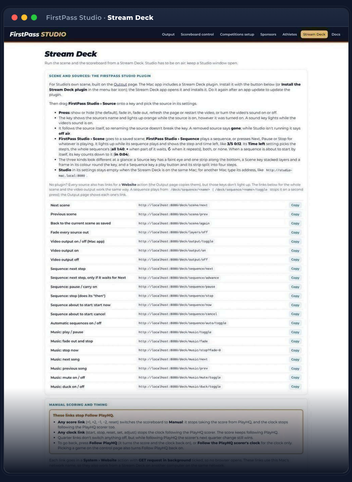


FIG 9.1

The Stream Deck tab: the plugin, the scoring and clock links, and the full reference below them.

9.1 The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Four kinds of key, each a different shape at a glance:

KEY	LOOKS	PRESS	SHOWS
Source	A faint eye and one strip along the bottom	Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off	The source's name; lit in its output's colour while it's on, however it was turned on
Scene	Stacked layers and a frame in its colour	Go to a scene, next or previous scene, or back to the current scene as saved	Lit while its scene is on, with a yellow dot when sources have changed since
Sequence	A play button and a strip in four steps	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself
Switcher	A camera and a strip in two halves	A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key, Fade to black, run a macro	The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air; a yellow dot while the switcher isn't answering

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**. Colour groups keys by the part of the stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads, or the scene's or sequence's own colour.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). A lit key takes its output's colour (Output 1 orange, 2 pink, 3 blue, 4 green) with a black title, so a commentators' Stream Deck of Commentary keys lights pink while the stream's keys light orange. A key for a removed output says **no output**.

> **Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them.**
>
> CHAPTER 9 · STREAM DECK

9.2 Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Stream Deck tab copy them, with this Mac's Stream Deck key after the port (`…:8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Stream Deck tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices…** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices… can take that one away without touching your other keys. The useful ones:

GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.
Switcher	Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro.

A LAYOUT THAT WORKS

Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

9.3 Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.

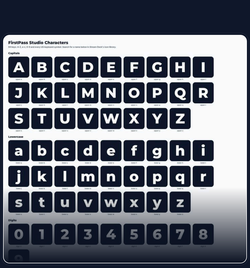


FIG 9.2 The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.



CHAPTER 10 OF 11

Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

1 The 1P menu · 2 The data folder · 3 Settings in iCloud Drive · 4 Backup and restore · 5 Updates · 6 Links and other devices · 7 When something isn't right · 8 Data and bandwidth

10.1 The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices…**, **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up…** and **Restore…**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Only this Mac can change things**, **What's new…** and Quit.

10.2 The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

10.3 Settings in iCloud Drive

Settings folder > Move to iCloud Drive… moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac…** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac…** undoes the move. Studio goes off air for a moment for each and comes straight back.

10.4 Backup and restore

Back up… saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore…** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

10.5 Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

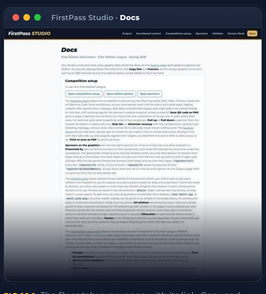


FIG 10.1 The Docs tab; every page with its link, Copy and Preview, the page options, and the notes this guide draws on.

> **Updates download by themselves and install when the app quits, never mid-stream.**
>
> CHAPTER 10 · SETTINGS AND CARE

10.6 Links and other devices

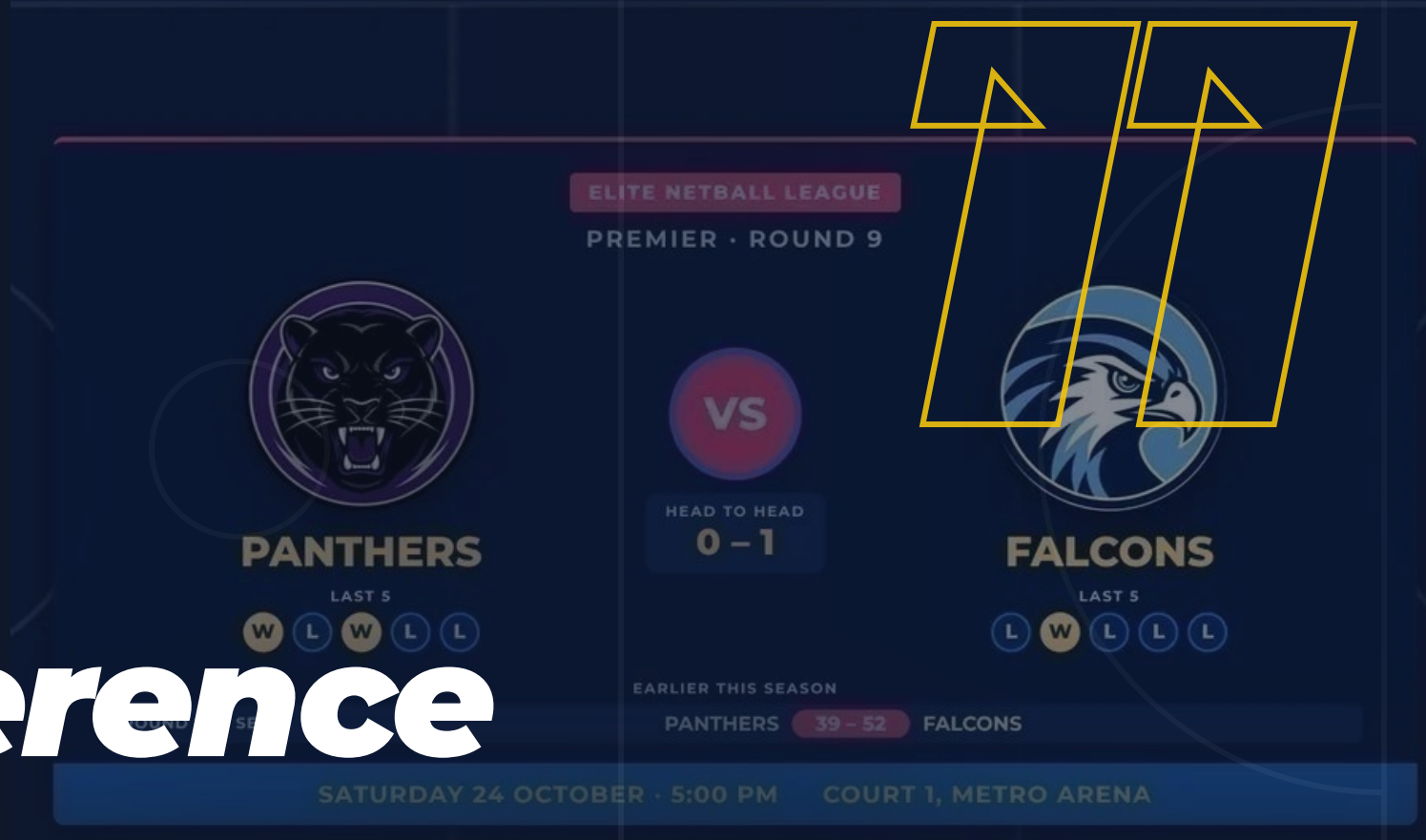
Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings. **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device works with the box ticked, and **Devices…** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

10.7 When something isn't right

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering. Switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong night or game is on	Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

10.8 Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game (or as set, down to one a second), the ticker's other courts every 30 seconds, the fixture about once a minute for automatic selection, and the rest once when a graphic loads. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth.



CHAPTER 11 OF 11

Quick reference

1 Keyboard
2 Stream night in nine lines
3 Where things are set

11.1 Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘9 go to Output, Production Planner, Scoreboard control, Competitions setup, Sponsors, Athletes, Stream Deck, Socials, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

11.2 Stream night in nine lines

01
Open Studio from the 1P menu and keep the Output page open all night.

02
Check the competition in use and the court on Competitions setup.

03
On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.

04
Turn on the video output your switcher takes and show one graphic to see it land.

05
Play a few seconds of music and an animation with sound.

06
Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.

07
Go to the Live game scene and let the sequences run the breaks.

08
Pick finals winners as they happen; turn on lower thirds when people appear.

09
Press Stream close, turn the output off, and send the sponsor report.

11.3 Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, mics, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Each night's court, the commentators and who calls each game	Production Planner
Keys and links	Stream Deck
Pictures of a game night for social media	Socials
Backups, iCloud, updates, the debug log	The 1P menu

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

FirstPass STUDIO

MADE BY PIVT

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.